

Macaulay's contribution [372] was, essentially, the equivalence (i) $\iff$ (ii) of Theorem 8.34. We have combined it with important work by Stanley [512, 513] and by Björner, Frankl & Stanley [93].

The sequences characterized in Theorem 8.34 are called *M-sequences* (dial “M” for “Macaulay”) or “*O*-sequences” (“O” for whatever). They are of fundamental importance, as we will also see in the next section.

Here are a few examples of *M*-sequences, for $d = 3$. The sequence $(1, 3, 3, h_3)$ is an *M*-sequence for $0 \leq h_3 \leq 4$. In fact, part (ii) of Macaulay's theorem suggests that for this we can take the multicomplex

$$\{ \emptyset, 1, 2, 3, 11, 12, 22 \}$$

together with the first h_3 sets from the list

$$F_1(3) = 111, \quad F_2(3) = 112, \quad F_3(3) = 122, \quad F_4(3) = 222.$$

Since $F_5(3) = 113$ contains the submultiset 13 that is not among the 1-faces we listed, we get that $(1, 3, 3, 5)$ is no longer an *M*-sequence. Note that among these, for $h_3 = 1$ we get $(1, 3, 3, 1)$ as an *M*-sequence: this is the h -vector of the boundary of an octahedron; see Example 8.22.

8.6 The g -Theorem and Its Consequences

From the last section, I hope we gathered some intuition for “what an *M*-sequence is.” All kinds of interpretations are useful: so, the best is to alternate between various explanations, between

- the f -vector of a multicomplex,
- the h -vector of a shellable complex, and
- a sequence of nonnegative integers satisfying $\partial^k(h_k) \leq h_{k-1}$.

Here comes one big reason why *M*-sequences are useful. It yields a complete characterization of the f -vectors of simplicial d -polytopes P . What do we know about them so far? Forgetting about $f_d = 1$, we know they can be encoded by their h -vectors $\mathbf{h}(P) = (h_0, h_1, \dots, h_d)$, which satisfy the Dehn-Sommerville equations $h_k = h_{d-k}$ for $0 \leq k \leq d$. Also, we know that $\mathbf{h}(P)$ is an *M*-sequence from Macaulay's Theorem 8.34(iv), which implies the upper bound inequality

$$h_k \leq \left(\binom{h_1}{k} \right) = \binom{h_1 + k - 1}{k}.$$

In quite a daring step, McMullen in 1970 combined all the then available information (including the lower bound theorem by Barnette, see below)